Introduction

This assignment focuses on using Bash scripting and Linux command-line tools to automate common system administration tasks. The project consists of two scripts that automate user environment management and directory backups.

Task 1 is a user environment automation script. It reads user information from a CSV file and automates the creation and configuration of Linux user accounts. This includes setting user passwords, configuring secondary groups, creating shared folders, applying appropriate permissions, creating symbolic links, and configuring a custom myls alias for sudo users.

Task 2 is a directory backup script. It accepts a local directory, creates a gzip-compressed tar archive of the directory, verifies the archive, and uploads the backup to a remote server using SCP over SSH. The script also verifies that the uploaded backup exists on the remote server and records important execution information in a log file.

Both scripts were tested during development, including successful executions and deliberately invalid inputs to demonstrate their error-handling functionality.

SCREENSHOTS:

```
student@lambcd1-host:~/ID616001-Assignment1/task1$ cat users.csv
email,birth_date,groups,shared_folder
alice.smith@example.com,1990/06/15,sudo:dev,/opt/staffData
bob.jones@example.com,1985/11/02,dev,/opt/staffData
carol.white@example.com,2000/03/22,intern,/opt/internData
dave.nguyen@example.com,1978/09/30,dev:docker,/opt/staffData
student@lambcd1-host:~/ID616001-Assignment1/task1$ _
```

Initial state, the users.csv file downloaded from the op.bit url and saved for local use

```
student@lambcd1-host:~/ID616001-Assignment1/task1$ ./create_users.sh https://osc.op-bit.nz/share/users.csv_
```

Command for executing the task 1 script (was pushed off screen by output so added separately)

```
CSV header:
email,birth_date,groups,shared_folder
CSV header is valid

Processing user:
        Email: alice.smith@example.com
        Birth Date: 1990/06/15
        Groups: sudo:dev
        Shared folder: /opt/staffData
        Email format IS valid
        Birth date format IS valid
        Username: sAlice
        Default password: 199006
        Username available - can create user

Processing user:
        Email: bob.jones@example.com
        Birth Date: 1985/11/02
        Groups: dev
        Shared folder: /opt/staffData
        Email format IS valid
        Birth date format IS valid
        Username: jBob
        Default password: 198511
        Username available - can create user

Processing user:
        Email: carol.white@example.com
        Birth Date: 2000/03/22
        Groups: intern
        Shared folder: /opt/internData
        Email format IS valid
        Birth date format IS valid
        Username: wCarol
        Default password: 200003
        Username available - can create user

Processing user:
        Email: dave.nguyen@example.com
        Birth Date: 1978/09/30
        Groups: dev:docker
        Shared folder: /opt/staffData
        Email format IS valid
        Birth date format IS valid
        Username: nDave
        Default password: 197809
        Username available - can create user

Number of users to be added: 4
Continue with user creation? (y/n): _
```

Initial output of task 1 script.

```
Number of users to be added: 4
Continue with user creation? (y/n): n
User creation cancelled
student@lambcd1-host:~/ID616001-Assignment1/task1$
```

Cancelling script at that y/n with n

```
Number of users to be added: 4
Continue with user creation? (y/n): y
User creation confirmed

Creating user: sAlice
[sudo: authenticate] Password:
```

When we select y, we are prompted to enter the sudo password. This is because we are assigning sudo privilege to this user (based on groups above)

```
Creating user: sAlice
[sudo: authenticate] Password:
        User created successfully
        Default password set successfully
        Password change at first login enabled
        Group already exists: sudo
        User added to group: sudo
        Group created: dev
        User added to group: dev
        myls alias created for sudo user
        Shared folder created: /opt/staffData
        Shared folder group set to: dev
        Shared folder permissions set to 770
        Shared link created: /home/sAlice/shared -> /opt/staffData

Creating user: jBob
        User created successfully
        Default password set successfully
        Password change at first login enabled
        Group already exists: dev
        User added to group: dev
        Shared folder already exists: /opt/staffData
        Shared folder group set to: dev
        Shared folder permissions set to 770
        Shared link created: /home/jBob/shared -> /opt/staffData

Creating user: wCarol
        User created successfully
        Default password set successfully
        Password change at first login enabled
        Group created: intern
        User added to group: intern
        Shared folder created: /opt/internData
        Shared folder group set to: intern
        Shared folder permissions set to 770
        Shared link created: /home/wCarol/shared -> /opt/internData

Creating user: nDave
        User created successfully
        Default password set successfully
        Password change at first login enabled
        Group already exists: dev
        User added to group: dev
        Group created: docker
        User added to group: docker
        Shared folder already exists: /opt/staffData
        Shared folder group set to: dev
        Shared folder permissions set to 770
        Shared link created: /home/nDave/shared -> /opt/staffData
student@lambcd1-host:~/ID616001-Assignment1/task1$ _
```

Users successfully created from the users.csv file. Default passwords based on birth date shown earlier are set using chage.

```
student@lambcd1-host:~/ID616001-Assignment1/task1$ getent passwd sAlice jBob wCarol nDave
sAlice:x:1003:1004::/home/sAlice:/bin/bash
jBob:x:1004:1006::/home/jBob:/bin/bash
wCarol:x:1005:1007::/home/wCarol:/bin/bash
nDave:x:1006:1009::/home/nDave:/bin/bash
student@lambcd1-host:~/ID616001-Assignment1/task1$
```

Command showing the successfully created users.

```
student@lambcd1-host:~/ID616001-Assignment1/task1$ getent group sudo dev intern docker
sudo:x:27:student,frodo,samwise,sAlice
dev:x:1005:sAlice,jBob,nDave
intern:x:1008:wCarol
docker:x:1010:nDave
student@lambcd1-host:~/ID616001-Assignment1/task1$ _
```

Command showing successfully created groups

```
student@lambcd1-host:~/ID616001-Assignment1/task1$ id sAlice
uid=1003(sAlice) gid=1004(sAlice) groups=1004(sAlice),27(sudo),1005(dev)
student@lambcd1-host:~/ID616001-Assignment1/task1$ ls -ld /home/sAlice
drwxr-x--- 2 sAlice sAlice 4096 Sep 11 01:21 /home/sAlice
student@lambcd1-host:~/ID616001-Assignment1/task1$
```

Individual account configuration SS 1 - showing created user details

```
Ubuntu 26.04 LTS lambcd1-host tty1

Try contacting this VM's SSH server via 'ssh vsock%246313318' from host.

lambcd1-host login: sAlice
Password:
You are required to change your password immediately (administrator enforced).
Changing password for sAlice.
Current password:
New password:
Retype new password:
The password has not been changed.
New password:
Retype new password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-31-generic x86_64)
```

Individual account configuration SS 2 – changing password on new users first login (forced by chage in script). Displays incorrect input responses until complete.

(Note: This produced an error a couple of times about authentication token manipulation, but eventually worked. I believe the problem has to do with VM state rather than the code, but this is a potential problem for code defense.)

```
student@lambcd1-host:~$ ls -ld /opt/staffData /opt/internData
drwxrwx--- 2 root intern 4096 Sep 11 01:44 /opt/internData
drwxrwx--- 2 root dev    4096 Sep 11 01:44 /opt/staffData
student@lambcd1-host:~$
```

Shared Folder group permissions are both 770. Dev and intern access accordingly.

```
student@lambcd1-host:~/ID616001-Assignment1/task1$ sudo find /home -maxdepth 2 -type l -name shared -ls
   440284      0 lrwxrwxrwx   1 root     root           14 Sep 11 01:44 /home/sAlice/shared -> /opt/staffData
   440294      0 lrwxrwxrwx   1 root     root           15 Sep 11 01:44 /home/wCarol/shared -> /opt/internData
   440289      0 lrwxrwxrwx   1 root     root           14 Sep 11 01:44 /home/jBob/shared -> /opt/staffData
   440299      0 lrwxrwxrwx   1 root     root           14 Sep 11 01:44 /home/nDave/shared -> /opt/staffData
student@lambcd1-host:~/ID616001-Assignment1/task1$
```

The Shared Symbolic Links of each user / group

```
sAlice@lambcd1-host:~$ myls
total 32
drwxr-x--- 3 sAlice sAlice 4096 Sep 11 01:51 .
drwxr-xr-x 9 root   root   4096 Sep 11 01:44 ..
-rw-r--r-- 1 sAlice sAlice   22 Sep 11 01:44 .bash_aliases
-rw------- 1 sAlice sAlice    7 Sep 11 01:51 .bash_history
-rw-r--r-- 1 sAlice sAlice  220 Feb 13  2026 .bash_logout
-rw-r--r-- 1 sAlice sAlice 3771 Feb 13  2026 .bashrc
drwx------ 2 sAlice sAlice 4096 Sep 11 01:48 .cache
-rw-r--r-- 1 sAlice sAlice  807 Feb 13  2026 .profile
lrwxrwxrwx 1 root   root     14 Sep 11 01:44 shared -> /opt/staffData
sAlice@lambcd1-host:~$ _
```

The alias for a created user

```
[2026-09-11 01:44:16] INFO: Username available: nDave
[2026-09-11 01:44:16] INFO: Number of users to be added: 4
[2026-09-11 01:44:19] INFO: User creation confirmed
[2026-09-11 01:44:19] Creating user: sAlice
[2026-09-11 01:44:19] SUCCESS: User created: sAlice
[2026-09-11 01:44:19] SUCCESS: Default password set for: sAlice
[2026-09-11 01:44:19] SUCCESS: First-login password change enabled for: sAlice
[2026-09-11 01:44:19] INFO: Group already exists: sudo
[2026-09-11 01:44:19] SUCCESS: sAlice added to group: sudo
[2026-09-11 01:44:19] SUCCESS: Group created: dev
[2026-09-11 01:44:19] SUCCESS: sAlice added to group: dev
[2026-09-11 01:44:19] SUCCESS: myls alias created for sudo user: sAlice
[2026-09-11 01:44:19] SUCCESS: Shared folder created: /opt/staffData
[2026-09-11 01:44:19] SUCCESS: Shared folder group set to dev: /opt/staffData
[2026-09-11 01:44:19] SUCCESS: Shared folder permissions set to 770: /opt/staffData
[2026-09-11 01:44:19] SUCCESS: Shared link created: /home/sAlice/shared -> /opt/staffData
[2026-09-11 01:44:19] Creating user: jBob
[2026-09-11 01:44:19] SUCCESS: User created: jBob
[2026-09-11 01:44:19] SUCCESS: Default password set for: jBob
[2026-09-11 01:44:19] SUCCESS: First-login password change enabled for: jBob
[2026-09-11 01:44:19] INFO: Group already exists: dev
[2026-09-11 01:44:19] SUCCESS: jBob added to group: dev
[2026-09-11 01:44:19] INFO: Shared folder already exists: /opt/staffData
[2026-09-11 01:44:19] SUCCESS: Shared folder group set to dev: /opt/staffData
[2026-09-11 01:44:19] SUCCESS: Shared folder permissions set to 770: /opt/staffData
[2026-09-11 01:44:19] SUCCESS: Shared link created: /home/jBob/shared -> /opt/staffData
[2026-09-11 01:44:19] Creating user: wCarol
[2026-09-11 01:44:19] SUCCESS: User created: wCarol
[2026-09-11 01:44:19] SUCCESS: Default password set for: wCarol
[2026-09-11 01:44:19] SUCCESS: First-login password change enabled for: wCarol
[2026-09-11 01:44:19] SUCCESS: Group created: intern
[2026-09-11 01:44:19] SUCCESS: wCarol added to group: intern
[2026-09-11 01:44:19] SUCCESS: Shared folder created: /opt/internData
[2026-09-11 01:44:19] SUCCESS: Shared folder group set to intern: /opt/internData
[2026-09-11 01:44:19] SUCCESS: Shared folder permissions set to 770: /opt/internData
[2026-09-11 01:44:19] SUCCESS: Shared link created: /home/wCarol/shared -> /opt/internData
[2026-09-11 01:44:20] Creating user: nDave
[2026-09-11 01:44:20] SUCCESS: User created: nDave
[2026-09-11 01:44:20] SUCCESS: Default password set for: nDave
[2026-09-11 01:44:20] SUCCESS: First-login password change enabled for: nDave
[2026-09-11 01:44:20] INFO: Group already exists: dev
[2026-09-11 01:44:20] SUCCESS: nDave added to group: dev
[2026-09-11 01:44:20] SUCCESS: Group created: docker
[2026-09-11 01:44:20] SUCCESS: nDave added to group: docker
[2026-09-11 01:44:20] INFO: Shared folder already exists: /opt/staffData
[2026-09-11 01:44:20] SUCCESS: Shared folder group set to dev: /opt/staffData
[2026-09-11 01:44:20] SUCCESS: Shared folder permissions set to 770: /opt/staffData
[2026-09-11 01:44:20] SUCCESS: Shared link created: /home/nDave/shared -> /opt/staffData
[2026-09-11 01:44:20] Script completed successfully
```

Task 1 timestamped log file output

Task 2:

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ ./backup.sh
Task 2 - Directory Backup Script

Enter the directory to backup: ../task1
Input received: ../task1
Source directory exists
Source directory: /home/student/ID616001-Assignment1/task1
Directory name: task1

Creating backup:
        backup-2026-09-11.tar.gz
Backup created successfully
Backup file: /home/student/ID616001-Assignment1/task2/backup-2026-09-11.tar.gz
Backup size: 8.0K

Testing backup archive
Backup archive IS valid

Backup contents:
task1/
task1/create_users.sh
task1/cleanup.sh
task1/users.csv
task1/test_users.csv
task1/create_users.log

Remote backup destination
Enter remote IP address or hostname: 192.168.100.129
Enter SSH port (press Enter for 22):
Enter remote username: kali
Enter remote destination directory: /var/assessments/task2

Remote host: 192.168.100.129
Remote port: 22
Remote user: kali
Remote directory: /var/assessments/task2

Checking remote connection...
kali@192.168.100.129's password:
```

Backup script SS 1 – choosing folder to backup, checking it exists, creating and testing, show contents, input destination details, repeat those details back to user

```
Checking remote connection...
kali@192.168.100.129's password:
Remote connection successful

Checking remote destination directory...
kali@192.168.100.129's password:
Remote destination directory exists

Uploading backup...
kali@192.168.100.129's password:
backup-2026-09-11.tar.gz
Backup uploaded successfully

Verifying remote backup...
kali@192.168.100.129's password:
Remote backup verified successfully

========================================
BACKUP COMPLETE
========================================
Source: /home/student/ID616001-Assignment1/task1
Backup: backup-2026-09-11.tar.gz
Remote: kali@kali@192.168.100.129:/var/assessments/task2

student@lambcd1-host:~/ID616001-Assignment1/task2$
```

Backup script SS 2 - continues from previous, entering destination user password for check connection and directory, uploading and verifying. Report complete when complete with details.

This .tar.gz file is also the output of the SCP command we used in the script (not rsync)

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ ssh kali@192.168.100.129 "ls -l /var/assessments/task2/"
kali@192.168.100.129's password:
total 8
-rw-rw-r-- 1 kali kali 6987 Sep 10 22:09 backup-2026-09-11.tar.gz
student@lambcd1-host:~/ID616001-Assignment1/task2$ _
```

Tunneling into the destination VM to show the .tar.gz file landed in the expected destination directory

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ cat backup.log
[2026-09-11 00:38:20] Script started
[2026-09-11 00:38:20] Input received: ../task1
[2026-09-11 00:38:20] SUCCESS: Source directory exists: ../task1
[2026-09-11 00:38:20] Source directory confirmed: /home/student/ID616001-Assignment1/task1
[2026-09-11 00:38:20] Creating backup: backup-2026-09-11.tar.gz
[2026-09-11 00:38:20] SUCCESS: Backup created: /home/student/ID616001-Assignment1/task2/backup-2026-09-11.tar.gz
[2026-09-11 00:38:20] Backup size: 8.0K
[2026-09-11 00:38:20] SUCCESS: Backup archive passed integrity test
[2026-09-11 00:38:20] SUCCESS: Backup contents displayed
[2026-09-11 00:40:11] Remote destination: kali@192.168.100.129:/var/assessments/task2
[2026-09-11 00:40:11] Remote SSH port: 22
[2026-09-11 00:40:14] SUCCESS: Remote SSH connection established
[2026-09-11 00:40:16] SUCCESS: Remote destination exists: /var/assessments/task2
[2026-09-11 00:40:16] Uploading backup: backup-2026-09-11.tar.gz
[2026-09-11 00:40:18] SUCCESS: Backup uploaded to kali@192.168.100.129:/var/assessments/task2
[2026-09-11 00:40:21] SUCCESS: Remote backup verified: /var/assessments/task2/backup-2026-09-11.tar.gz
[2026-09-11 00:40:21] Script completed successfully
[2026-09-11 02:07:22] Script started
[2026-09-11 02:07:47] Input received: ../task1
[2026-09-11 02:07:47] SUCCESS: Source directory exists: ../task1
[2026-09-11 02:07:47] Source directory confirmed: /home/student/ID616001-Assignment1/task1
[2026-09-11 02:07:47] Creating backup: backup-2026-09-11.tar.gz
[2026-09-11 02:07:47] SUCCESS: Backup created: /home/student/ID616001-Assignment1/task2/backup-2026-09-11.tar.gz
[2026-09-11 02:07:47] Backup size: 8.0K
[2026-09-11 02:07:47] SUCCESS: Backup archive passed integrity test
[2026-09-11 02:07:47] SUCCESS: Backup contents displayed
[2026-09-11 02:08:35] Remote destination: kali@192.168.100.129:/var/assessments/task2
[2026-09-11 02:08:35] Remote SSH port: 22
[2026-09-11 02:09:14] SUCCESS: Remote SSH connection established
[2026-09-11 02:09:19] SUCCESS: Remote destination exists: /var/assessments/task2
[2026-09-11 02:09:19] Uploading backup: backup-2026-09-11.tar.gz
[2026-09-11 02:09:21] SUCCESS: Backup uploaded to kali@192.168.100.129:/var/assessments/task2
[2026-09-11 02:09:25] SUCCESS: Remote backup verified: /var/assessments/task2/backup-2026-09-11.tar.gz
[2026-09-11 02:09:25] Script completed successfully
student@lambcd1-host:~/ID616001-Assignment1/task2$
```

The output of the task 2 script logging file.

INTENTIONAL ERROR SCREENSHOTS:

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ ./backup.sh /does/not/exist
Task 2 - Directory Backup Script

Input received: /does/not/exist
ERROR: Directory does not exist: /does/not/exist
student@lambcd1-host:~/ID616001-Assignment1/task2$
```

Running script on nonexistent file

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ ./backup.sh
Task 2 - Directory Backup Script

Enter the directory to backup: ../task1
Input received: ../task1
Source directory exists
Source directory: /home/student/ID616001-Assignment1/task1
Directory name: task1

Creating backup:
        backup-2026-09-11.tar.gz
Backup created successfully
Backup file: /home/student/ID616001-Assignment1/task2/backup-2026-09-11.tar.gz
Backup size: 8.0K

Testing backup archive
Backup archive IS valid

Backup contents:
task1/
task1/create_users.sh
task1/cleanup.sh
task1/users.csv
task1/test_users.csv
task1/create_users.log

Remote backup destination
Enter remote IP address or hostname: 192.0.2.1
Enter SSH port (press Enter for 22):
Enter remote username: kali
Enter remote destination directory: /var/assessments/task2

Remote host: 192.0.2.1
Remote port: 22
Remote user: kali
Remote directory: /var/assessments/task2

Checking remote connection...
ssh: connect to host 192.0.2.1 port 22: Connection timed out
ERROR: Unable to connect to remote host
student@lambcd1-host:~/ID616001-Assignment1/task2$ _
```

Unreachable remote link error

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ ./backup.sh ../task1
Task 2 - Directory Backup Script

Input received: ../task1
Source directory exists
Source directory: /home/student/ID616001-Assignment1/task1
Directory name: task1

Creating backup:
        backup-2026-09-11.tar.gz
Backup created successfully
Backup file: /home/student/ID616001-Assignment1/task2/backup-2026-09-11.tar.gz
Backup size: 8.0K

Testing backup archive
Backup archive IS valid

Backup contents:
task1/
task1/create_users.sh
task1/cleanup.sh
task1/users.csv
task1/test_users.csv
task1/create_users.log

Remote backup destination
Enter remote IP address or hostname: 192.168.100.129
Enter SSH port (press Enter for 22):
Enter remote username: wronguser
Enter remote destination directory: /var/assessments/task2

Remote host: 192.168.100.129
Remote port: 22
Remote user: wronguser
Remote directory: /var/assessments/task2

Checking remote connection...
wronguser@192.168.100.129's password:
Permission denied, please try again.
wronguser@192.168.100.129's password:
Permission denied, please try again.
wronguser@192.168.100.129's password:
wronguser@192.168.100.129: Permission denied (publickey,password).
ERROR: Unable to connect to remote host
student@lambcd1-host:~/ID616001-Assignment1/task2$ _
```

Incorrect username / password output

```
student@lambcd1-host:~/ID616001-Assignment1/task2$ ./backup.sh ../task1
Task 2 - Directory Backup Script

Input received: ../task1
Source directory exists
Source directory: /home/student/ID616001-Assignment1/task1
Directory name: task1

Creating backup:
        backup-2026-09-11.tar.gz
Backup created successfully
Backup file: /home/student/ID616001-Assignment1/task2/backup-2026-09-11.tar.gz
Backup size: 8.0K

Testing backup archive
Backup archive IS valid

Backup contents:
task1/
task1/create_users.sh
task1/cleanup.sh
task1/users.csv
task1/test_users.csv
task1/create_users.log

Remote backup destination
Enter remote IP address or hostname: 192.168.100.129
Enter SSH port (press Enter for 22):
Enter remote username: kali
Enter remote destination directory: /does/notexist

Remote host: 192.168.100.129
Remote port: 22
Remote user: kali
Remote directory: /does/notexist

Checking remote connection...
kali@192.168.100.129's password:
Remote connection successful

Checking remote destination directory...
kali@192.168.100.129's password:
ERROR: Remote destination directory does not exist or is inaccessible
student@lambcd1-host:~/ID616001-Assignment1/task2$
```

Directory does not exist error